

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (CCE) (Credit System) Degree Examinations**

January 2023

20CC501 - WIRELESS NETWORKS AND MOBILE COMPUTING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Elaborate the differences between wired and wireless communications.	6	L*2	1	1
b)	Explain the principle of an Ad Hoc network and list their advantages and limitations.	7	L2	1	1
c)	Explain the concept of hidden nodes problem in wireless networks.	7	L2	1	1
2. a)	Explain the role of interframe spacing in 802.11 MAC and draw the framing diagram.	8	L2	1	1
b)	Is energy consumption a critical aspect for wireless devices? Explain with suitable example.	6	L2	1	1
c)	Discuss the error recovery process with DCF.	6	L2	1	1
Unit – II					
3. a)	Explain in detail two types of authentication schemes used in 802.11.	10	L2	2	1
b)	Define scanning. Explain different types of scanning with neat diagram.	10	L2	2	1
4. a)	Discuss the role of Authentication algorithm number, beacon interval, Transaction sequence number and capability information.	10	L2	2	1
b)	Discuss the 802.11 generic management frame with frame diagram.	10	L2	2	1
Unit – III					
5. a)	What are the features proposed for 4G system?	8	L2	3	1
b)	Discuss the Location Management scheme for location area with example.	8	L2	3	1
c)	Discuss the applications of mobile computing.	4	L2	3	1
6. a)	Discuss the anatomy of 5G Samsung mobile devices.	6	L2	3	1
b)	Discuss the types of handoff between the MS and BS.	6	L2	3	1
c)	Discuss the tunnelling and registration process in mobile IP.	8	L2	3	1
Unit – IV					
7. a)	Explain the termination detection using Distributed snapshot algorithm.	7	L2	4	1
b)	Explain the Message-optimal termination detection algorithm with example.	7	L2	4	1
c)	Discuss the effect of cache in mobile computing.	6	L2	4	1
8. a)	Point out different types of Data Delivery Mechanism. Explain any one of them with their advantages and disadvantages.	10	L2	4	1
b)	Explain the Spanning Tree based termination detection algorithm with example.	10	L2	4	1
Unit – V					
9. a)	Discuss the Security issues in mobile environment.	10	L2	5	1
b)	Discuss the requirements of mobile agent system.	10	L2	5	1
10. a)	What is cloud computing? Explain the essential characteristics of it.	10	L2	5	1
b)	Discuss the transaction processing in mobile computing.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
